Cloud Computing Lab 5 – Assignment 1

AWS RDS PostgreSQL Integration with Redmine on EC2

Student	Santo Xavier
Roll No.	10699
Class	T.E. Computer Engineering – COMPS B
Subject	Cloud Computing
Academic Year	2026–2027

1. Objective

The objective of this assignment is to deploy an AWS RDS PostgreSQL database and connect it to the Redmine application deployed on the Lab 4 Amazon EC2 instance. The implementation demonstrates secure EC2-to-RDS connectivity and all mandatory CRUD operations.

2. Architecture

The following architecture shows the client-to-EC2-to-RDS flow. Redmine/Puma runs inside the EC2 instance, while PostgreSQL is hosted on a private RDS database subnet. RDS inbound access is restricted to the EC2 Security Group on TCP port 5432.

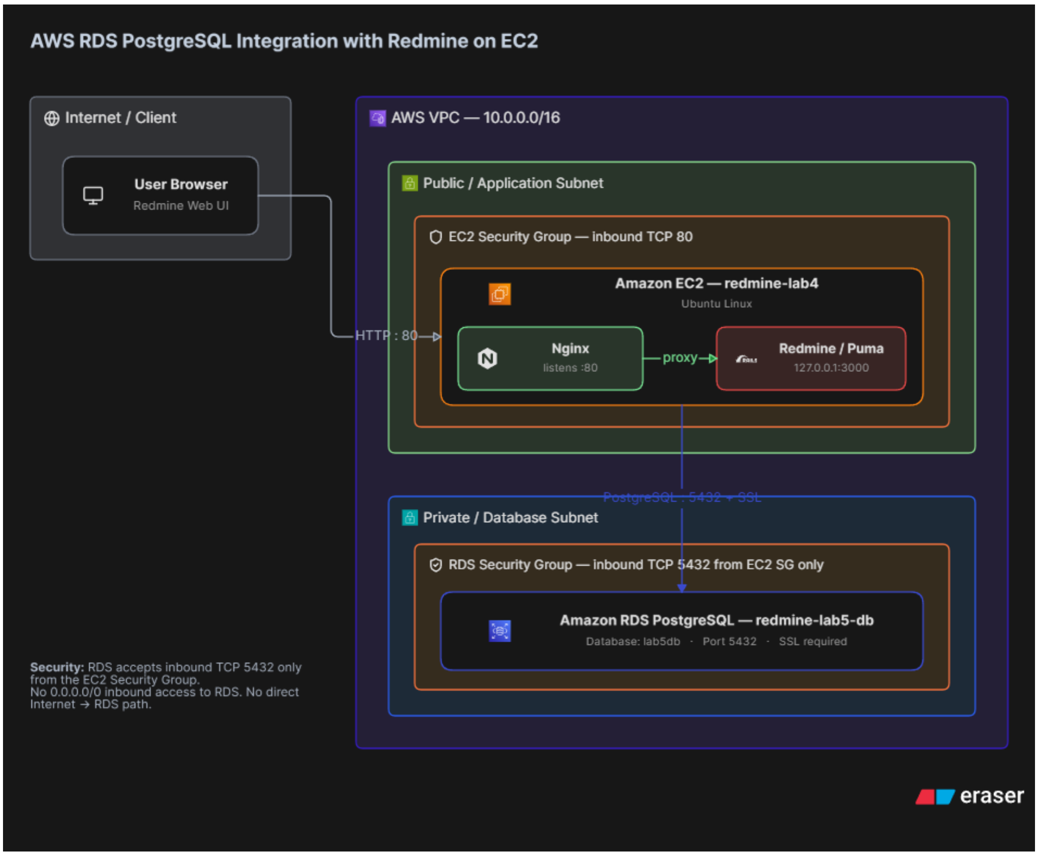

Figure 1: AWS RDS PostgreSQL and Redmine on EC2 architecture.

3. AWS RDS Deployment

An Amazon RDS PostgreSQL instance named **redmine-lab5-db** was deployed for the Lab 5 assignment. The database is named **lab5db** and uses PostgreSQL on port **5432**.

4. RDS Security Configuration

A dedicated RDS Security Group, **rds-lab5-sg**, was configured. Its inbound PostgreSQL rule permits TCP port 5432 only from the EC2 Security Group. No 0.0.0.0/0 inbound rule is configured for PostgreSQL, preventing direct public database access.

5. EC2 to RDS Connection

The existing Lab 4 Redmine application runs on the EC2 instance **redmine-lab4**. Nginx receives HTTP traffic on port 80 and forwards requests to Redmine/Puma on **127.0.0.1:3000**. Redmine connects to the RDS PostgreSQL database over PostgreSQL port 5432 using the configured EC2-to-RDS security relationship.

6. CRUD Demonstration

Operation	Evidence
Create	A new Redmine record/issue was created successfully.
Read	The created record was retrieved and displayed.
Update	The existing record was modified successfully.
Delete	The record was deleted successfully.

7. Application and Repository

EC2 Application URL: **http://13.63.171.224/**
GitHub Repository: **https://github.com/SAnto-spec/cloud-computing-lab5-rds-redmine-10699**

8. Conclusion

AWS RDS PostgreSQL was successfully deployed and integrated with the Redmine application running on the Lab 4 EC2 instance. Database access was secured by allowing PostgreSQL traffic only from the EC2 Security Group. The application demonstrated the mandatory Create, Read, Update and Delete operations.